CAP Proposal

2025/2026 Second Term

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The project topic selection

The topic selection of our prototype is an Aquaponics system.

Imagine growing fresh, crisp lettuce and plump, healthy fish in your own home, without soil, without the need for endless watering, and without harsh chemicals. Sounds like a dream? That's exactly what aquaponics makes possible.

Aquaponics is a revolutionary approach to gardening that combines aquaculture (raising fish) with hydroponics (growing plants in water). The result is a self-sustaining ecosystem where your fish feed on your plants, and your plants in turn, keep your fish tank clean. It's efficient, eco-friendly, and perfect for anyone interested in sustainable gardening.

Unlike traditional soil gardening, aquaponics can save up to 90% of water and doesn't require chemical fertilizers, making it an ideal option for eco-conscious gardeners.

The Benefits of Aquaponics

1. Grow Fresh Food Year-Round

No matter the season, aquaponics systems allow you to harvest fresh vegetables, herbs, and fish all year long. Unlike soil gardening, you're not limited by the climate outside.

2. Water Efficiency

Aquaponics uses up to 90% less water than traditional gardening. The water in your system recirculates continuously, making it perfect for drought-prone areas or anyone looking to conserve resources.

3. Eco-Friendly and Chemical- Free

With no need for chemical fertilizers or pesticides, aquaponics is a natural, sustainable way to grow food. Your plants thrive, your fish thrive, and you're contributing to a healthier planet.

How Aquaponics Differs from Other Growing Methods

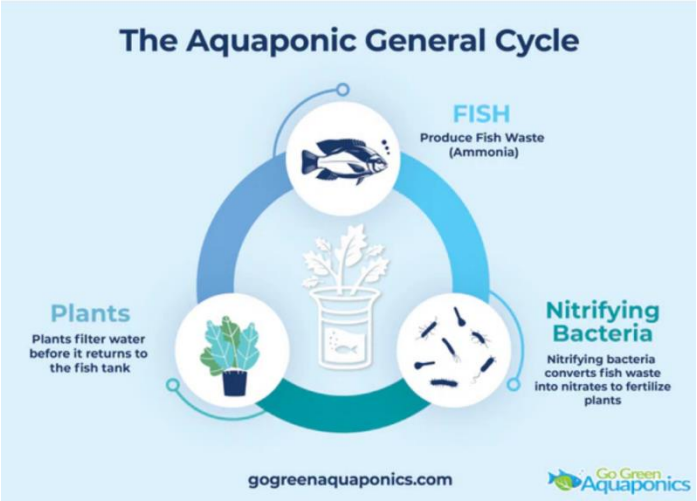
1. Aquaponics vs. Hydroponics

While both aquaponics and hydroponics grow plants without soil, hydroponics relies on adding synthetic nutrient solutions to water. In contrast, aquaponics uses natural fish waste as a fertilizer, making it more organic and sustainable. There's no need to buy bottled nutrients, and the system doesn't produce wastewater.

2. Aquaponics vs. Traditional Soil Gardening

Unlike soil gardening, aquaponics uses up to 90% less water, eliminates weeding, and can be done indoors or in compact spaces. There's no soil erosion, no runoff, and fewer pest issues. Plus, it produces two sources of food, plants and fish, from one system.

The simple mechanism of the Aquaponics



The problem to be solved

Figure 1 shows the challenges we face.

Rapid population growth, increasing food demand, and global water scarcity have intensified pressure on conventional agricultural and aquaculture systems, **as shown in Figure** 1. Agriculture alone accounts for approximately 70% of global freshwater withdrawals, much of which is lost through inefficient irrigation methods such as evaporation, runoff, and uneven water distribution. At the same time, aquaculture activities generate nutrient-rich wastewater that is often discharged untreated, contributing to environmental pollution despite containing valuable nutrients.

Aquaponics has emerged as a sustainable alternative that integrates aquaculture and hydroponic plant production within a closed-loop system. Research indicates that aquaponics can reduce water consumption by up to 80–90% compared to traditional soil-based agriculture by continuously recirculating water. In this system, fish waste serves as a natural nutrient source for plants, while plants act as biological filters that remove excess nutrients, improving water quality before it returns to the fish tanks.

By recycling water and nutrients, aquaponics minimizes wastewater discharge, reduces reliance on chemical fertilizers, and supports efficient food production. This dual-output system produces both plant crops and protein-rich fish, addressing food security challenges while conserving limited water resources. Therefore, aquaponics represents a promising, research-backed solution to the interconnected problems of water waste, environmental pollution, and sustainable food production.

Evidence:

https://doi.org/10.21608/pserj.2024.252618.1291

https://doi.org/10.21608/agro.2024.282754.1424

(PDF) Aquaponics system for sustainable water, energy, and food nexus: A review

Methodology

How Does Aquaponics Work?

While it may sound complex, aquaponics is just a carefully balanced ecosystem. Here's a beginner-friendly breakdown of how it all works:

Step 1: Fish Provide Nutrients

Fish produce waste in the water, which contains nitrogen and other nutrients essential for plant growth.

Step 2: Bacteria Convert Waste into Plant Food

Beneficial bacteria naturally break down fish waste into forms that plants can absorb. Think of them as your microscopic garden helpers!

Step 3: Plants Filter the Water

As plants take in nutrients, they clean the water, which is then returned to the fish tank, creating a self-sustaining loop.

There are many types of aquaponics, but our type is:

Raft System of Deep-Water Culture (DWC)

Also known as raft aquaponics, DWC systems float plants on a sheet of foam over deep troughs filled with nutrient-rich water, as shown in Figure 2. The roots dangle directly into the water, absorbing nutrients continuously.

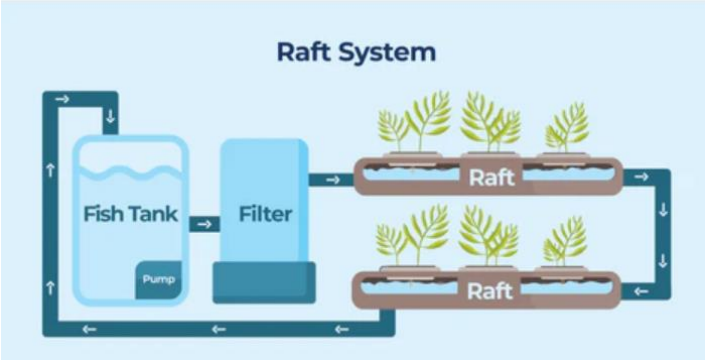


Figure 2 shows the shape of the DWC

How it works:

- Fish waste is filtered out before reaching the grow bed.
- Clean, oxygenated water flows through the raft bed.
- Air stones or diffusers are used to oxygenate the water to prevent root rot.

Benefits:

- Highly scalable for commercial systems
- Ideal for fast-growing crops like lettuce, kale, basil, and chard
- Lower maintenance once established

Evidence: https://gogreenaquaponics.com/blogs/news/what-is-aquaponics-and-how-does-it-work

How the wastes of fish are converted into the food of plants

In floating-raft aquaponic systems, fish waste is first released into the water primarily as ammonium (NH_4^+). This ammonium is toxic for fish if it accumulates, so it is biologically converted into forms usable by plants through a process called nitrification. First, ammonia-oxidizing bacteria such as *Nitrosomonas* convert ammonium to nitrite (NO_2^-):

$$2NH_4^+ + 3O_2 \rightarrow 2NO_2^- + 2H_2O + 4H^+$$

Next, nitrite-oxidizing bacteria like *Nitrobacter* oxidize nitrite to nitrate (NO_3^-), which is the primary plant-available nitrogen form:

$$2NO_2^- + O_2 \rightarrow 2NO_3^-$$

This process allows the plants on the rafts to absorb nitrate as a nutrient, while the water is naturally purified and recirculated back to the fish tank.

Evidence: https://doi.org/10.1016/j.ibiod.2017.08.006

The type of fish that we will use

Not all fish are suited for aquaponics. Some species are too sensitive, others don't produce enough nutrients, and some require conditions that are difficult to maintain without specialized equipment. When the wrong fish is chosen, problems like poor plant growth, stressed fish, unstable water chemistry, and system crashes become far more likely**.**

The Best Fish for Aquaponics
Aquaponics enthusiasts often consider several key fish species for their systems because of their specific advantages and considerations. Here's a detailed look at some of the top choices:

Tilapia

Ideal pH Range: 6.5 - 9.
Ideal Temperature Range: 82 - 86 °F
Ideal Fish Tank Size: A fully grown tilapia will weigh 1 pound, although they can grow larger. The rule of thumb is that one pound of tilapia will need 3 gallons of water. This means that in every 3 to 5 gallons of water, only one tilapia.
Temperature Requirements: Ensure the water temperature is suitable for tilapia, generally between 75-85°F (24-30°C).

Figure 3 shows Tilapia

Advantages of tilapia in aquaponics:

1. **Rapid Growth:** Tilapia are known for their fast growth rate, making them an excellent choice for aquaponics. They reach harvestable size quickly so you can enjoy a consistent supply of fish.
2. **High Reproduction Rate:** They reproduce readily, helping maintain a steady fish population.

3. Wide Temperature Tolerance: Tilapia can adapt to various water temperatures, making them suitable for various climates.
4. **Hardiness**: They are hardy fish that can tolerate fluctuations in water quality, making them forgiving for beginners.
5. **Mild Taste**: Tilapia have a mild, white flesh that is palatable to most consumers, making them a popular choice for aquaponic growers.

Considerations of raising tilapia in aquaponics:

1. **Dietary Needs**: Tilapia are primarily herbivorous but can be omnivorous to some extent.
2. **Warm Water**: While they can adapt to varying temperatures, tilapia thrive in warmer waters. In cooler climates, you may need heating systems to maintain ideal conditions.
3. **Breeding**: Tilapia can reproduce prolifically, which can lead to overpopulation if not managed carefully

Evidence: https://gogreenaquaponics.com/blogs/news/what-are-the-best-fish-for-aquaponics

What Plants Grow Best in a Deep-Water Culture System?

Leafy greens and herb**s** are the undisputed champions of DWC.

These plants have relatively low oxygen demands, grow quickly in water, and are perfectly suited for the constant moisture of a DWC system.

1- **Lettuce:** All types excel—Butterhead, Romaine, Oakleaf, and Loose-leaf. They grow incredibly fast and rarely bolt (go to seed) in a controlled environment

What is Hydroponic Lettuce?

Growing Hydroponic Lettuce is lettuce cultivated without soil, using a nutrient-rich water solution. This method allows plants to absorb minerals directly through their roots, often resulting in faster growth, higher yields, and efficient resource use. It's ideal for indoor gardening due to its compact nature and minimal space requirements.

Figure 4 shows the lettuce.

Benefits of Hydroponic Lettuce

- Year-round growth, faster harvests (~30% quicker than soil).
- Water-efficient (up to 90% less water).
- No soil-borne pests, often pesticide-free.

2. Herbs (Flavor Powerhouses)

Herbs grown in DWC are often more aromatic and potent than their soil-grown counterparts.

- **Basil:** The king of DWC herbs. It grows into a massive, bushy plant. Genovese, Thai, and Lemon basil all perform exceptionally well.
- **Evidence:** https://www.betilife.com/blogs/hydroponics/how-to-grow-hydroponic-lettuce-indoors-at-home
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What is hydroponic Basil?

Hydroponic Basil is basil (Ocimum basilicum) grown without soil, using a nutrient-rich water solution instead. Its roots are either suspended directly in the solution or supported by an inert medium like rockwool or clay pebbles. It's cultivated in controlled environments where water, nutrients, light, and climate are precisely managed, potentially leading to faster growth, higher yields, and year-round availability of fresh basil plants.

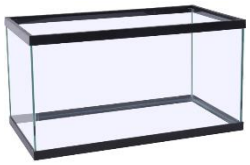
Hydroponic Basil Benefits & Key Features

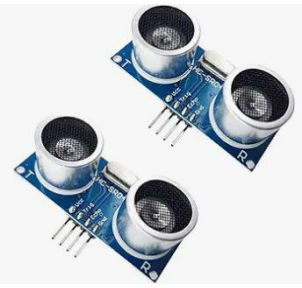
- **Faster Growth & Higher Yields:** Plants get optimal nutrients and oxygen directly to their roots, leading to faster growth cycles and potentially higher harvests per square foot compared to soil.

- **Water Efficiency:** Hydroponic systems recirculate water, using up to 90% less water than traditional soil farming.
- **Controlled Environment:** Grown indoors (greenhouses, vertical farms, grow rooms) or in protected settings, allowing for:
    - **Year-Round Production:** Independent of seasons or weather.
    - **Pesticide/Herbicide-Free:** The controlled environment drastically reduces pest and disease pressure, often eliminating the need for chemical pesticides or herbicides.
    - **Consistent Quality & Flavor:** Precise control over nutrients and environment leads to uniform size, appearance, and taste.
- **Reduced Risk of Soil-Borne Diseases:** No soil means no exposure to common soil pathogens.
- **Space Efficiency:** Vertical stacking (vertical farming) maximizes production in small areas.
- **Cleaner Harvest:** Plants and leaves aren't covered in soil, making them cleaner at harvest.

Evidence: https://www.betilife.com/blogs/hydroponics/grow-hydroponic-basil

Materials

The name	The usage	The shape
Fish tank	To carry the water And the fish	
Flexible Tube	Transfer the water from and to the fish tank and the Raft Grow bed	
Raft Grow Bed	To carry the plants and water with neutrons	
Water pump	To pump the water from and to the fish tank and the Raft Grow bed	
Air pump	Pumping the air for the fish	
DC fan	For cooling the water if its temperature is below the permitted level for fish.	

Ultrasonic sensor	Measure the water level in the fish tank automatically.	
DS18B20 Temperature Sensor		
Solenoid Valve	controlling water flow automatically	
Arduino uno	The head of the connections	
Lettuce	Filter the water from the wastes of the fish as food	

Light intensity sensor	Detect the light intensity of the plant	
LED Grow Lights	Providing the plant with the light	
Tilapia	Producer of the plant food	

The parameters and actuators that will be used

1-Temperature

- Water temperature is measured using a **DS18B20 waterproof temperature sensor**, which provides real-time temperature readings from the fish tank.

Figure 5 shows ***DS18B20***

Actuator:

- Water heater
- DC fan

Figure 7 DC fan

Figure 6 water heater

2-water level

Water level is measured using an **ultrasonic sensor**, which calculates the distance between the sensor and the water surface to determine the water level inside the fish tank.

Figure 8 ***ultrasonic sensor***

Actuator

- Solenoid valve
- Water pump

Figure 9 water pump

Figure 10 solenoid valve

3-light intensity

Light intensity is measured using a **digital light sensor (BH1750)**, which provides accurate illumination values in lux.

Figure 11 ***digital light sensor (BH1750),***

Actuator

- LED grow light

Figure 12 LED grow light

For the microcontroller, the Arduino Uno is the best one for the solution. Which have

Arduino Uno	Processor	Memory	Digital I/O	Analogue I/O
Arduino Uno	16Mhz ATmega328	2KB SRAM, 32KB flash	14	6 input, 0 output

Eventually, this controller will control each sensor to gather data, process it, and compare it with the plants' standards.

And dealing with any deviation in system readings by actuators

Figure 13 Arduino Uno

The creative part of our project

We will use a generator after the plant growth bed to convert the mechanical action into electric action

Benefits

It will provide an amount of electricity for our project

renewable system

Real-life application

We chose Borg El Arab City in Alexandria government to be our real-life location due to its environmental and demographic characteristics that contrast with our school's location. Unlike the arid desert climate of October Gardens, Borg El Arab features a coastal environment with an average humidity of 70% and a moderate, stable climate year-round. This selection allows us to address the challenge of increasing the industrial and agricultural base of Egypt. Furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egypt Grand Challenge of dealing with population growth and its consequences. The city provides a broad land availability for scalable agriculture, serving as a strategic hub to increase Egypt's agricultural base.

Safety Considerations

To ensure the safety of students and prevent damage to the aquaponics system, all electrical components are kept away from water and properly insulated. Students must turn off the power supply before performing any maintenance or adjustments. Gloves should be worn when handling water, pipes, or electrical parts to avoid contamination and injury. The system should be regularly checked for leaks, loose connections, and damaged wires to prevent water loss or electrical hazards. Proper supervision is required during operation to ensure safe handling and system stability.